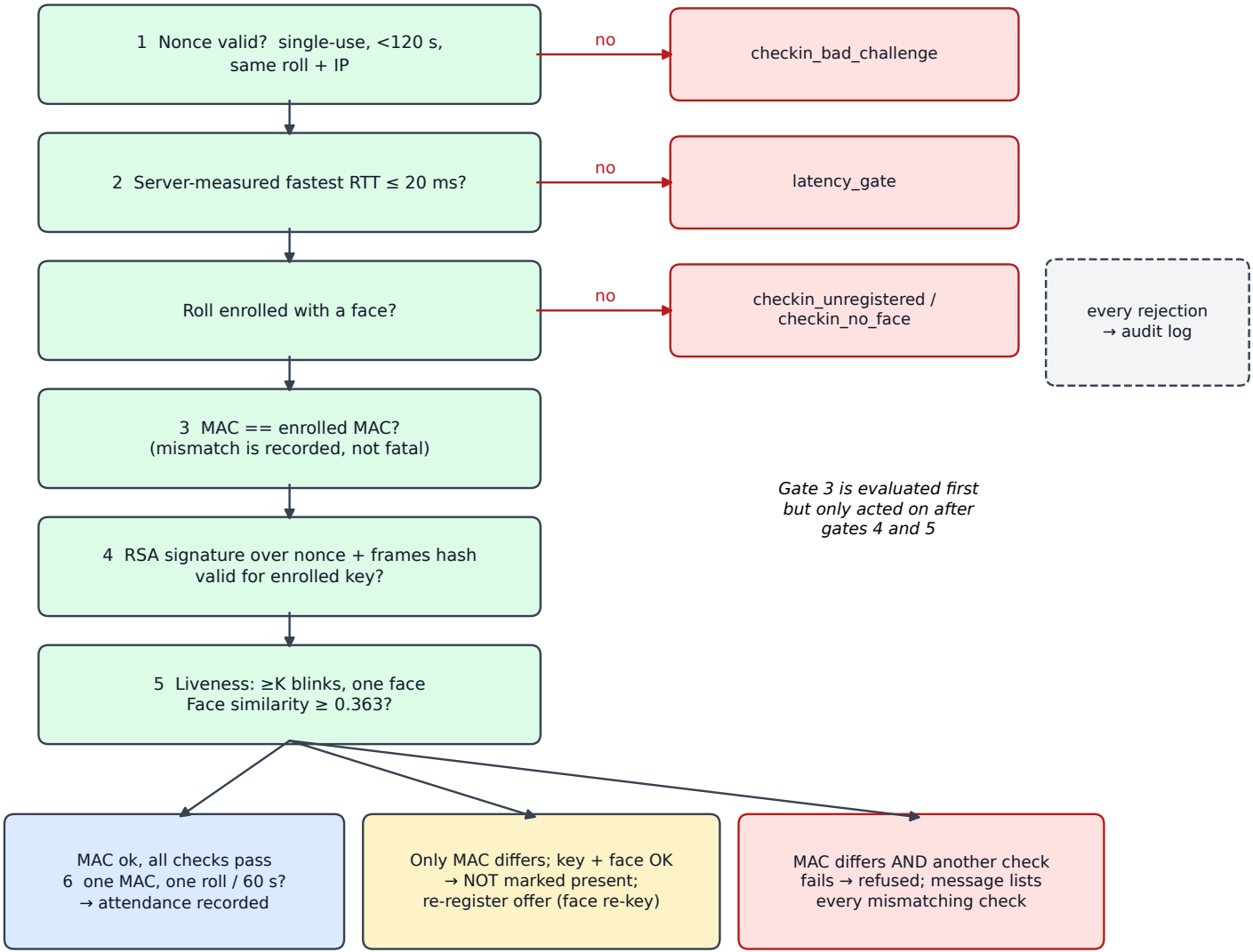
1 Nonce valid? single-use, <120 s, same roll + IP
no
checkin_bad_challenge
2 Server-measured fastest RTT ≤ 20 ms?
no
latency_gate
Roll enrolled with a face?
no
checkin_unregistered / checkin_no_face
every rejection → audit log
3 MAC == enrolled MAC? (mismatch is recorded, not fatal)
Gate 3 is evaluated first but only acted on after gates 4 and 5
4 RSA signature over nonce + frames hash valid for enrolled key?
5 Liveness: ≥K blinks, one face Face similarity ≥ 0.363?
MAC ok, all checks pass 6 one MAC, one roll / 60 s? → attendance recorded
Only MAC differs; key + face OK → NOT marked present; re-register offer (face re-key)
MAC differs AND another check fails → refused; message lists every mismatching check